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Smart Scheduling System

Graduation Project 2

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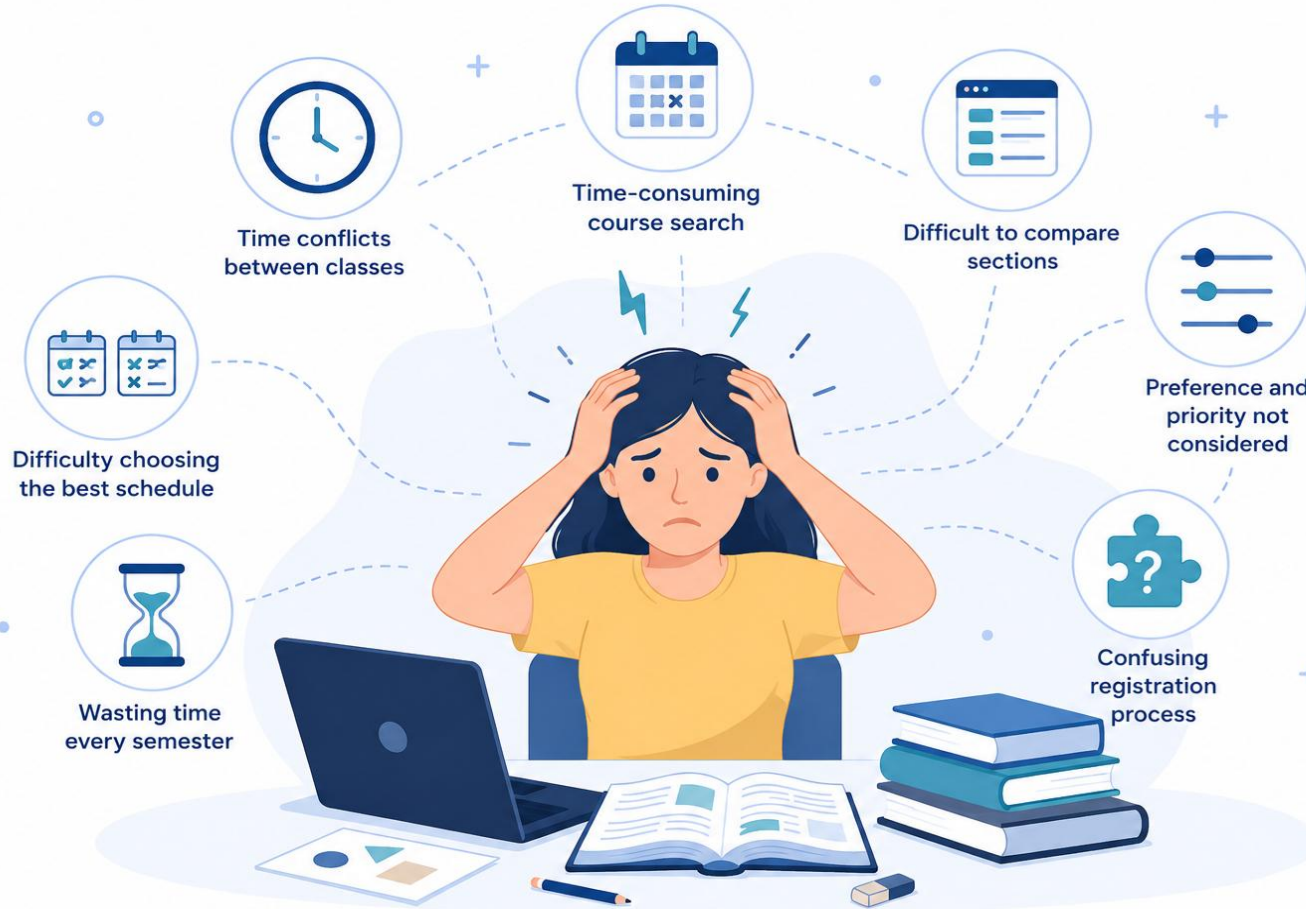
Introduction



During every registration period, students spend significant time comparing sections, avoiding schedule conflicts, and searching for suitable class times and instructors.

This process can be stressful, repetitive, and time-consuming.

Problem Statement



SOLUTION



Step 1

Select Courses:

Choose the required courses for registration.

Step 2

Set Preferences:

Select preferred days, time range, breaks, and instructors.

Step 3

Generate Schedules:

The system creates multiple clash-free schedule options automatically.

Step 4

Compare Options:

Review and compare generated schedules.

Step 5

Save or Export:

Save favorite schedules or download them as PDF.

Key Features

Powerful tools to create the perfect schedule — intelligently.



BUILD YOUR PREFERENCES



Conflict-Free
Generation



Day
Preferences



Time
Range



REFINE YOUR SCHEDULE



Credit
Hours



Break
Hours



Instructor
Preferences



SMARTER RESULTS



AI Schedule
Ranking



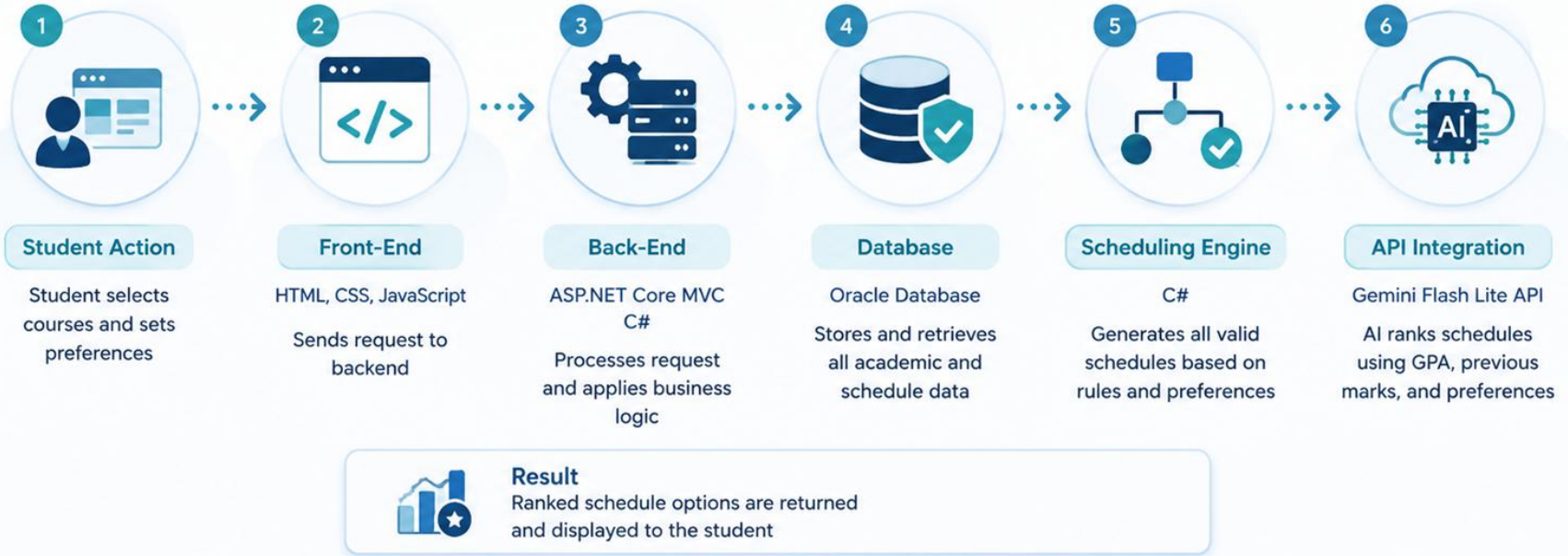
Favorite
Schedules



PDF
Export

System Architecture Flow

From student request to intelligent schedule results



Tech Stack



Front-End
HTML, CSS, JS



Back-End
ASP.NET Core MVC
C#



Database
Oracle DB



Engine
C#



AI API
Gemini Flash
Lite API

Schedule Generation Algorithm



Generating all valid conflict-free schedules automatically



Purpose

Generate all valid conflict-free schedules automatically based on:



Selected courses



Student preferences



Credit hour constraints



Time conflict prevention

1

Input
Selected
Courses



2

Generate
Section
Combinations



3

Check
Time
Conflicts



4

Check
Credit
Hours



5

Check
Prerequisites



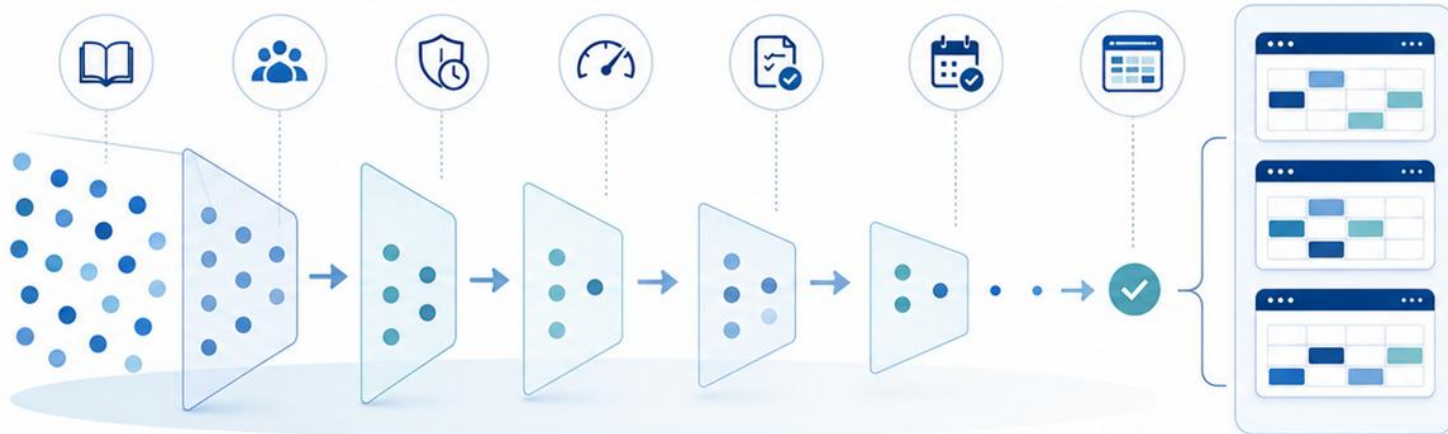
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Add Valid
Schedules



7

Return
Valid
Schedules



Simplified Pseudocode

- 1 Input selected courses
- 2 Generate all possible section combinations
- 3 For each combination:
- 4 Return all valid schedules



Check
time conflicts



Check
credit hours



Check
prerequisites

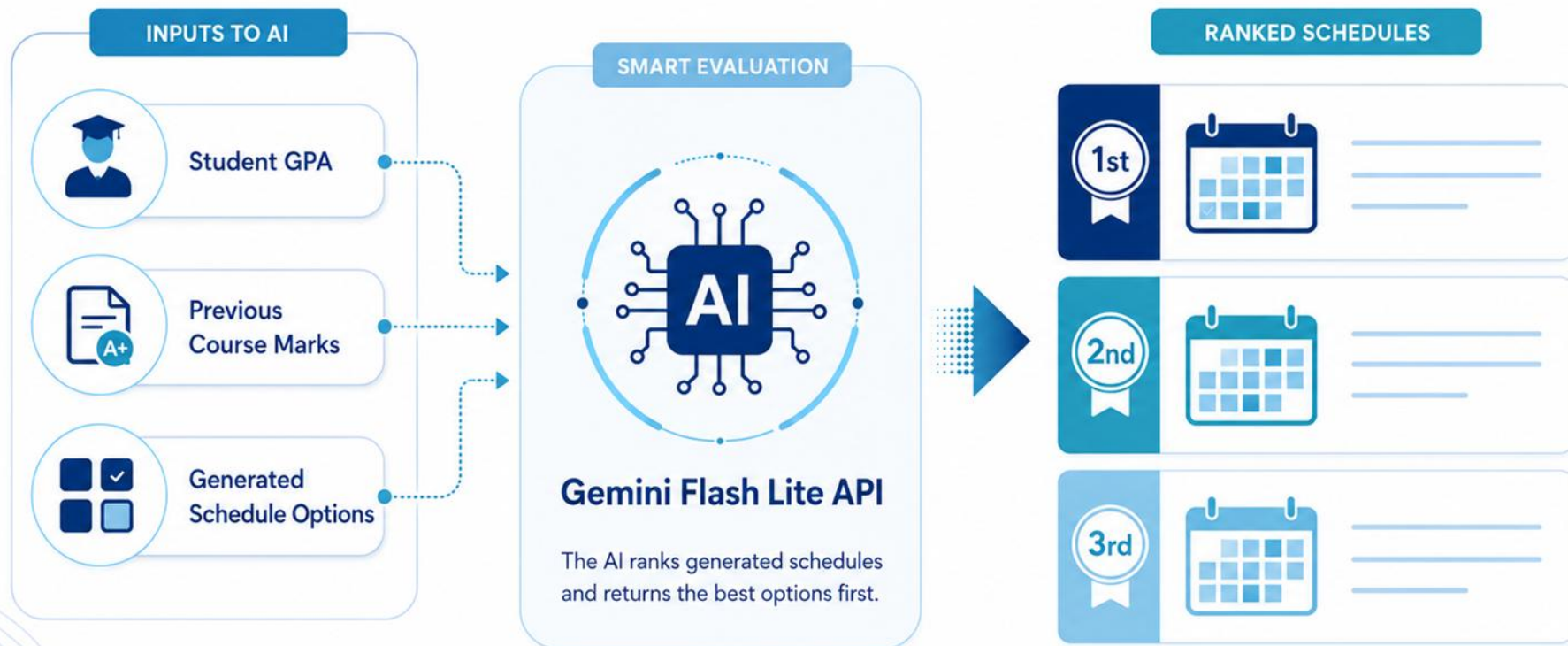


If valid:
add to valid
schedules list



AI Schedule Ranking

Ranking schedules from best to least suitable



Testing & Validation Approach

Verifying the main system workflow and generated schedule results



Conclusion



The Smart Student Schedule System helps students generate personalized, conflict-free schedules quickly and easily.

By reducing manual effort and supporting student preferences, the system makes the registration experience simpler, faster, and more organized.



Thank You!